

Module 4: Engineering Economic Analysis

1. You plan to buy a second hand 5 door hatchback car. You have two choices:

- A Nissan Leaf electric vehicle (2022, 10,000 km, \$25,000)
- A Toyota Corolla petrol vehicle (2022, 10,000 km, \$20,000)

Your annual mileage is 10,000 km. You estimate you will own the car for 15 years, and at the end of that time you will scrap it.

- The current price of petrol is 240 cents per litre (cpl), and the Corolla uses 6 litres/100 km.
- The Leaf uses 20 kWh/100 km, and charges at night at a discounted rate of 17c/kWh.

Petrol and electricity prices are forecast to increase at a rate of 2.50% per annum for the next 15 years.

You currently have money on deposit at the bank at 4%, and you plan to use that money to fund the installation.

You mention this purchase to your bank manager who suggests a rule of thumb that you not invest in any proposal with a payback greater than a quarter of its nominal life.

Ignoring all tax implications,

- a) Calculate the time it takes to payback the extra capital cost.
- b) Calculate the net present value of both options.
- c) Which option should you go with, and why?
- d) What should you say to the bank manager regarding his payback rule of thumb?
- e) State three significant financial risks for your chosen option. How would you quantify each risk?

a. Leaf:

Annual electricity costs = 10,000 km * 20 kWh/100 km * 17 c/kWh = \$340/year.

Corolla:

Annual petrol costs = 10,000 km * 6 L/100 km * 240 c/L = \$1440/year.

Savings per annum = 1440 - 340 = \$1100 per year.

Payback = 5000/1100 = 4.5 years (Note that \$5000 is the increased capital cost for the leaf compared to the corolla).

b. Leaf:

$C_0 = -\$25,000$ (cost)

$C_1 - C_{15} = -\$340$ (without growth) - annuity.

$PV_{annuity, growing} = \frac{C}{r-g} [1 - (\frac{1+g}{1+r})^t] = \frac{-340}{0.04-0.025} [1 - (\frac{1.025}{1.04})^{15}] = -4,438.39$

NPV = $-\$29,438.39$

Corolla:

$C_0 = -\$20,000$ (cost)

$C_1 - C_{15} = -\$1440$ (without growth) - annuity.

$PV_{annuity, growing} = \frac{C}{r-g} [1 - (\frac{1+g}{1+r})^t] = \frac{-1440}{0.04-0.025} [1 - (\frac{1.025}{1.04})^{15}] = -18,797.90$

NPV = $-\$38,797.90$

- c. Go ahead with the Leaf, as its NPV is better than the Corolla's.
- d. Tell the bank manager that a payback of 4.5 years on a project with a 15 year lifespan may break their rule of thumb, but payback analysis is flawed because it does not take the time value of money into account, it favours early cash flows (ignoring cash flows after payback) and the determination of acceptable payback is arbitrary.
- e. Three possible risks are:
 - Change in cost increases - the 2.5% per annum increase in prices could change over the next 15 years. Could perform a sensitivity analysis to quantify the risk (vary the growth rates and see what impact that has on the NPVs).
 - Actual vehicle lifetime lower than expected - should research actual market lifetimes of similar vehicles to improve confidence in 15 year lifespan.

- Taxation - tax implications have been ignored, but could impact one or both options if tax is applied/reduced for either option. Test the model with a set of potential taxation impacts to estimate the potential impacts.

2. In 2023, 87% of New Zealand's electricity generation was from renewable sources. The National Policy Statement for Renewable Energy Generation has set at target of 90% by 2025.

Longitude Energy, a South Island electricity generator and retailer, believe that the simplest and most efficient way for New Zealand to achieve its renewable energy generation target is to build a new dam on the lower Waitaki River.

It is estimated that construction of the dam and power station will cost \$1,200 million. The levelised cost of electricity for hydroelectric generation is \$100/MWh, and currently Longitude Energy is charging its customers an average tariff of 25c/KWh. The design life of the dam is 30 years.

The NZ Treasury recommendation for analysis of energy infrastructure projects is a real discount rate of 6%. Longitude Energy has a company policy of investing only in projects with a payback less than or equal to three years.

- a) Assuming generation averages 2000 GWh/year calculate the payback and the NPV for the project. Ignore the effects of tax and inflation in your analysis.
- b) What advice would you give to Longitude Energy, and why? Justify that advice with reference to the relative strengths and weakness of the two methods.
- c) List two significant financial risks for the project.

a. LCOE is \$100/MWh

Revenue is 25c/kWh (\$250/MWh)

For 2000GWh, annual cash flow is $2000 \times 1000 \times 150 = \300 mill per annum

Payback is $1200/300 = 4$ years

NPV = $-1200 + \text{Annuity}$ (300 a year for 30 years at 6%)

= $-1200 + 300[\text{AF}]$ - the annuity factor from tables is 13.765 (or use formula for ordinary annuity)

= \$2,929.5 million

- b. Based on their payback criteria, Longitude would not invest. However, the project has a positive NPV, so it will make a positive contribution to the value of the company. Therefore your advice should be to undertake the project. The problems with Payback are that it does not take the time value of money into account, the selection of 3 years as an acceptable payback period is arbitrary, and it ignores cashflows that occurs after the payback period.

c. Two possible risks are:

- Change in electricity tariffs - increased competition, changes in technology or legislation could cause average tariffs to reduce, reducing income.
- Changes in generation - 2000 GWh/year modelled, but actual generation could be lower than this.

3. BigBang Energy is a new entrant into the New Zealand electricity market. They are planning to build a 50 MW geothermal plant.

The binary cycle geothermal system will cost \$85 million to build and \$6 million to operate a year. Construction of the plant will take two years, with construction costs evenly split over both years. Construction costs begin at the end of year 1. Operating costs do not begin until construction is finished. The plant is expected to last for 25 years after construction is complete. At that time, it is expected that the salvage value will be the same as the cost of removal and site remediation.

Electricity generated by the plant will sell for 7c/kWh, yielding an annual revenue of \$30 million.

If BigBang has an opportunity cost of 12%, determine the net present value of the geothermal plant. Ignore all tax effects.

$C_0 = \$0$

$C_1 = \$42.5$ million (cost)

$C_2 = \$42.5$ million (cost)

Costs C_3 - $C_{27} = \$6$ million p.a.

Costs are $PV(C_1) + PV(C_2) + PV(C_3-C_{27})$

$$\begin{aligned}
PV(C_1) &= \frac{42.5}{1.12} = \$37.95 \text{ million} \\
PV(C_2) &= \frac{42.5}{1.12^2} = \$33.88 \text{ million} \\
PV(C_3-C_{27}) &\text{ - annuity, deferred cashflow, so use difference of two annuities.} \\
PV(C_3-C_{27}) &= 6 * \left(\frac{1}{0.12} - \frac{1}{0.12(1+0.12)^{27}} \right) - 6 * \left(\frac{1}{0.12} - \frac{1}{0.12(1+0.12)^2} \right) = \$37.52 \text{ million} \\
PV_{\text{costs}} &= \$109.34 \text{ million} \\
\text{Revenue is } \$30 \text{ million p.a., starting in year 3 - annuity, deferred cashflow, so use difference of two annuities.} \\
PV_{\text{revenue}} &= 30 * \left(\frac{1}{0.12} - \frac{1}{0.12(1+0.12)^{27}} \right) - 30 * \left(\frac{1}{0.12} - \frac{1}{0.12(1+0.12)^2} \right) = \$187.58 \text{ million} \\
NPV &= PV_{\text{revenue}} - PV_{\text{costs}} = \$78.23 \text{ million.}
\end{aligned}$$

4. BigBang is considering expanding into the provision of sustainably produced lightweight battery technology for electric vehicles.

BigBang have been developing two prototype systems that offer similar functionality, although they are based on entirely different intellectual property. BigBang cannot afford to bring both methodologies to market and are relying on you to recommend which to progress with, based on an economic analysis of the net present values of both options.

Ignoring the effects of tax and inflation, calculate the net present value of both options and state which option you recommend. BigBang is using an opportunity cost of 7% for both options. All expenditures and revenues occur at the end of a year. Setup costs are paid at the end of year zero, and costs/revenue begin at the end of year 1.

The details of the two options are as follows:

Option 1

Option 1 has an initial setup cost of \$37 million dollars, and you predict that the ongoing operating and maintenance costs will be \$840,000 per annum for the foreseeable future. Revenue in the first year of operation will be \$1 million dollars, and this is likely to grow by 5% per annum indefinitely.

Option 2

Option 2 has an initial setup cost of \$20 million dollars, and likely operating and maintenance costs of \$500,000 per annum for the foreseeable future. Revenue will be consistent at \$2 million per annum with no growth for the foreseeable future.

Option 1

Costs: \$37 million now, plus \$840,000 perpetuity.

$$PV_{\text{perpetuity}} = \frac{C}{r}$$

$$PV_{\text{costs}} = \$37 \text{ million} + \frac{840,000}{0.07} = \$49 \text{ million.}$$

Revenue: perpetuity of \$ 1 million, 5% growth p.a. - growing perpetuity.

$$PV_{\text{revenue}} = \frac{1,000,000}{0.07-0.05} = \$50 \text{ million.}$$

$$NPV = \$1 \text{ million.}$$

Option 2

Costs: \$20 million now, plus \$500,000 perpetuity.

$$PV_{\text{costs}} = \$20 \text{ million} + \frac{500,000}{0.07} = \$27.14 \text{ million.}$$

Revenue: perpetuity of \$ 2 million.

$$PV_{\text{revenue}} = \frac{2,000,000}{0.07} = \$28.57 \text{ million.}$$

$$NPV = \$1.43 \text{ million.}$$

Go with option 2, as it has a higher NPV.

5. MacHack is a 3D printer produced by Contract Manufacturing Ltd (CML). MacHacks are assembled in NZ.

It takes 2.5 hours to assemble a unit. Labour costs \$20/hour, monthly Health and Safety training and compliance costs are \$150 per worker and materials costs are \$50/unit. Overheads per worker are \$2000/month. Each worker completes three MacHacks each day, filling the remainder of the day with overhead tasks, and works on average for 20 days a month.

- Determine the average cost of a unit in a production month.
- Assuming the proposed selling price of a MacHack is \$500/unit, how many must be sold per worker to break even each month?

- a. $VC = \$20/\text{hr} * 2.5 \text{ hr/unit} + \$50/\text{unit} = \$100/\text{unit}$.
 $FC/\text{month} = \$150 + \$2000 = \$2150/\text{month}$
 $\text{Total cost per month/worker} = (3 \text{ units per day} * 20 \text{ days per month}) * \$100/\text{unit} + \$2150/\text{mo/worker}$
 $= \$8150/\text{month}$.
 $\text{Average cost per unit per month} = \135.83
- b. Breakeven occurs when total revenue = total cost
 $\text{Breakeven number of units} = x$
 $\text{Selling price} * x = FC + VC * x$
 $500x = 2150 + 100x$
 $x = 5.375$ - Need to sell 6 units per worker per month to breakeven.

6. ACME Contract Manufacturing Ltd has one of its manufacturing plants operating on a single-shift five day work week producing specialised components for distributed computing systems. The plant is operating at its full capacity (24,000 units of output per week) without the use of overtime or shifts. Fixed costs for single-shift operation are \$90,000 per week. The average variable cost is \$30 per unit up to 24,000 units per week.

The company has received an order from Edison Electronics that will make total demand for 28,000 units per week. Two options are being considered to fill the new order:

Option 1: Increase the plant's output to up to 36,000 units a week by adding overtime, Saturday operations, or both. No increase in fixed costs is entailed, but the variable cost is \$36 per unit for any output in excess of 24,000 units per week, up to a 36,000 unit capacity.

Option 2: Operate a second shift. The maximum capacity of the second shift is 21,000 units per week. The variable cost of the second shift is \$31.50 per unit and its operation entails additional fixed costs of \$13,500 per week.

- a) What is the cheapest option to fulfil the order?
 - b) Determine the operating volume at which that option becomes the more profitable choice.
 - c) If each unit sells for \$100, what is the contribution margin per unit for the new order, using the cheapest option identified?
- a. The operating costs for the first shift operation ($FC = \$90,000$ $VC = 24,000 * \$30$) remain unchanged regardless of whether option 1 or 2 is chosen.
Therefore the cost for the extra 4000 units is:
Using option 1: $4000 * \$36 = \$144,000$
Using option 2: $4000 * \$31.5 + \$13,500 = \$139,500$
Option 2 is cheaper at 4000 units
 - b. We are looking for the break-even point for the additional volume over 24,000 units.
Option 1 additional $VC = \$36 * X$ (where X is the breakeven quantity)
Option 2 additional $VC = \$31.50 * X$, additional $FC = \$13,500$
Solving for X gives 3,000 units. Therefore option 2 is more profitable between 27,000 and 45,000 units per week. (Could show this graphically, but would have to be accurate and legible)
 - c. Using option 2, $VC = (24,000 * \$30) + (4,000 * \$31.5) = \$846,000$
 $CM = TR - VC = 100 * 28,000 - \$846,000 = \$1,954,000$
 $CM/\text{unit} = 1,954,000 / 28,000 = \$69.79/\text{unit}$

Module 5: Management and Financial Accounting

1. Explain the accounting equation.

The accounting equation states that:

Assets = Liabilities + Equity

This means that everything an entity owns (its assets) are paid for either by money the owners already have (equity) or by money they have borrowed (debt).

2. You are considering an offer to become a partner in Lux Energy Consultants, an engineering consultancy business. The most recent financial statements for Lux are shown in Fig. ??.
 - a) What does EBIT stand for, and what does it tell us? Calculate EBIT for Lux for the year ended March 31, 2023.
 - b) Calculate the current ratio for Lux. What does this tell you?
 - c) Calculate ROE. What does this tell you?
 - d) Based on the financial statements, what questions would you ask before you commit to becoming a partner?
 - a. EBIT stands for earnings before interest and tax. It tells us about the profitability of an entity before the effects of financing (tax and interest on debt) are taken into account. EBIT for Lux is \$199,000.
 - b. Current ratio = current assets/current liabilities = $30,000/30,000 = 1$.
A current ratio of one, while indicating that Lux is theoretically capable of paying off its debts, shows a company hovering on the brink of insolvency.
 - c. ROE = NPAT/total owner's equity = 9.8%.
The ROE for the company at 9.8% return is likely greater than money on deposit at the bank. At first glance this would seem to be a good investment. However is the extra return greater than the extra risk, given the current ratio?
 - d. Seeing as the company has \$200,000 tied up in a long term investment that returned only \$9,000, it would seem worthwhile investigating the release of that money early, and using to pay off some of the building mortgage (the rates are similar) and bolster the cash reserves and hence improve the current ratio. One might also ask, why was no tax paid? It would also be prudent to ask to see previous years' financial statements, to see what the historic trends are.

Lux Energy Consultants Statement of Financial Performance For year ending 31 March 2023		
Revenue		
Fees revenue	\$1,100,000	
Interest revenue	<u>\$9,000</u>	
Total revenue		<u>\$1,109,000</u>
<i>less</i> Expenses		
Building expenses	\$100,000	
Salaries/Wages	\$700,000	
Vehicle expenses	\$50,000	
Admin expenses	<u>\$60,000</u>	
Total expenses		<u>\$910,000</u>
<i>less</i> interest		\$22,725
<i>less</i> tax		\$0
Net Profit After Tax		<u>\$176,275</u>

Lux Energy Consultants Statement of Financial Position As at 31 March 2023		
Owner's Equity		
Capital	\$1,728,725	
<i>less</i> Drawings	\$110,000	
<i>plus</i> Net profit	<u>\$176,275</u>	<u>\$1,795,000</u>
Represented by:		
Assets		
Acc'ts receivable	\$15,000	
Cash on hand	\$15,000	
Investment (10 yr term deposit)	\$200,000	
Office building	\$2,000,000	
Vehicles	\$65,000	
Equipment	<u>\$30,000</u>	<u>\$2,325,000</u>
Less Liabilities		
Acc'ts payable	\$25,000	
Overdraft	\$5,000	
Mortgage (20 yr term)	<u>\$500,000</u>	<u>\$530,000</u>

Figure 1: Lux Energy Consultants financial statements

3. SaaS-R-Us is an innovative new software as a service company that is generating a great deal of media attention because of the commitment of its CEO to making SaaS-R-Us “the most fun workplace in the world”.

SaaS-R-Us senior management has also shared on social media that they are on the brink of launching cutting edge new software that will change the face of social media as we know it, and the current hype is huge.

They have recently had the intellectual property associated with the software valued at \$1,000,000.

You have been offered the opportunity to invest in SaaS-R-Us, and their Chief Financial Officer (CFO) has sent you the most recent financial statements, together with a promise of spectacular growth in the next eighteen months. These financial statements are shown in Fig. ??.

- a) Calculate EBIT for SaaS-R-Us for the year ended March 31, 2023.
- b) Calculate the current ratio for SaaS-R-Us. What does this tell you about the ongoing operation of the company?
- c) Calculate the debt-equity ratio for SaaS-R-Us. What does this tell you?
- d) Calculate ROE for SaaS-R-Us. Based on your calculation, should you invest in the company? Comment on your answer with respect to the saying “risk expects appropriate return”.
- e) Based on the financial statements, what are three questions you would ask before you commit to investing in SaaS-R-Us?

a. EBIT for SaaS-R-Us is $\text{income} - \text{total expenses} = \$3,281,399 - \$3,247,561 = \$33,838$.

b. Current ratio = $\text{current assets} / \text{current liabilities} = 394,87 / 30,479 = 12.96$

A current ratio of greater than 1 indicates that SaaS-R-Us is solvent and capable of paying off its short term debts.

c. Debt-equity ratio = $\text{total debt} / \text{total owner's equity} = 0.086$.

The debt-equity ratio for SaaS-R-Us is extremely low, which indicates that the company has taken on very little debt relative to investment. This means that the company has mostly been financed through equity investments, and could afford to take on debt if that could provide an economic advantage.

d. ROE = $\text{NPAT} / \text{total owner's equity} = 11,338 / 2,206,800 = 0.51\%$

Based on the current ROE, do not invest in SaaS-R-Us. The ROE for the company at 0.51% return is less than money on deposit at the bank. However, it is likely that investment in this company is riskier than depositing your money in the bank. Therefore investing does not make sense.

(We can assume that there is some risk associated with the launch of the new software, and that is reflected in the promise of spectacular growth – however growth in revenue does not equal growth in ROE, so we do not have enough information to say otherwise.

e. Some good questions would be:

- The company has \$400,000 tied up in a long term investment that returned only \$22,097 (5.5%). Why? It would seem worthwhile to investigate the release of some money early, in order to pay off some of the long term loan (\$160,279 at approx. 13%).
- There is a great deal of cash (\$218k) sitting around not earning – given the robust current ratio, this is unnecessary. What is the reason for it?
- There is a lot of short term debt outstanding (accounts receivable is \$175k). Why? Consider getting a debt collector in!
- One might also ask, why was no tax paid?
- What is the basis for the valuation of \$1 mill for intellectual property?
- Why has so much equity been issued, as opposed to debt taken on? The debt-equity ratio reveals there is significant room for more debt, rather than diluting investor returns (as evidenced by the low ROE).
- It would also be prudent to ask to see previous years' financial statements, to see what the historic trends are.

SaaS-R-U's	
Statement of Financial Position	
As at 31 March 2023	
Assets	
Acct's receivable	\$175,998
Cash on hand	\$218,873
Investment (10 yr term deposit)	\$400,000
Intellectual property	\$1,000,000
Vehicles	\$346,786
Equipment	<u>\$255,901</u>
Total Assets	\$2,397,558
Liabilities	
Acc'ts payable	\$25,479
Overdraft	\$5,000
Loan (20 yr term)	\$160,279
Total Liabilities	\$190,758
Equity	
Issued Capital	\$2,195,462
Retained Earnings	<u>\$11,338</u>
Total Equity	\$2,206,800

SaaS-R-U's	
Statement of Financial Performance	
Year Ended 31 March 2023	
Income	
Interest Income	\$22,097
<i>Sales Income:</i>	
Short term hire	\$494,732
Campaign services	\$686,668
Subscription services	\$2,077,902
Total Income	<u>\$3,281,399</u>
Operating Expenses	
Personnel	\$2,029,528
Development	\$15,300
Marketing	\$80,800
Operations (general overheads)	\$770,125
Operations (sales overhead)	\$351,808
Total Expenses	<u>\$3,247,561</u>
Interest	\$22,500
Tax	\$0
Net Profit After Tax	\$11,338

Figure 2: SaaS-R-U's financial statements